PSY 8960-003 Reproducibility Project: Examining the Association Between Neuroticism and Loneliness Among College Students

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The authors made the following contributions. Caroline Armstrong: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing.

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Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline. Two to three sentences of **more detailed background**, comprehensible to scientists in related disciplines. One sentence clearly stating the **general problem** being addressed by this particular study. One sentence summarizing the main result (with the words "**here we show**" or their equivalent). Two or three sentences explaining what the **main result** reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge. One or two sentences to put the results into a more **general context**. Two or three sentences to provide a **broader perspective**, readily comprehensible to a scientist in any discipline.

Keywords: keywords

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There is evidence that loneliness among emerging adults, including college students, has risen from the late 1970s to today (Buecker et al., 2021). A number of social changes have been suggested to be involved in this trend, including greater job instability and changes to communication patterns introduced by innovations in technology (Buecker et al., 2021). Examining such changes and the degree of evidence supporting their possible contributions to a "loneliness epidemic" (Killeen, 2002) is an important aim of research, such that interventions and policies might be designed to support young people's social connection and such that those supporting them in formal and informal roles (e.g., educators, parents, friends, coaches, and spiritual leaders) might be informed of potential contributors to loneliness and potential strategies to help mitigate them.

At the same time, it may be equally important to examine factors that tend to be comparatively more stable over time, including individual personality traits (Costa et al., 1986), as variables contributing to emerging adults' loneliness in and of themselves and as variables moderating the influence of other (e.g., social) variables on loneliness. One personality trait, neuroticism, is considered part of the five-factor model of personality (McCrae & John, 1992) and can generally be defined as a disposition "characterized by a chronic level of emotional instability and proneness to psychological distress" (APA, 2018). Neuroticism is hypothesized to be a vulnerability factor for anxiety and mood disorders (Barlow et al., 2014), in which loneliness is a central feature. Loneliness has also been shown to be a temporal antecedent of these disorders (e.g., Lim et al., 2016). These links as well as general characterizations—both lay and scientific—of neuroticism and loneliness suggest that the two may be closely interwoven. These demonstrated links to mental health outcomes also highlight the importance of examining each and taking steps to parse them conceptually. In fact, neuroticism and loneliness may be "sibling constructs" or perhaps cousin constructs, and empirically examining their correlations is important for describing their nomological networks and enhancing conceptual clarity in the fields that study these constructs (Lawson & Robins, 2021).

Some research has already taken such steps. One large meta-analysis ($k = 113$, total $n = 93,668$; Buecker et al., 2020) examined the connection between neuroticism and loneliness and found that they were positively correlated, r = .358. However, a large number of loneliness measures suffer from the limitation of asking about hypothesized causes of loneliness rather than "pure" loneliness. Accordingly, there is a need to examine the correlation between neuroticism and loneliness using a loneliness measure that merely asks one how lonely they feel. The Loneliness in Context Questionnaire (Asher & Weeks, 2013) was developed to address this shortcoming and is a promising instrument for just assessing loneliness itself, whatever an individual's interpretation of "lonely" is. There is also a need to examine this correlation, using this kind of loneliness measure, specifically within a sample of college students in order to better understand this association in this particular population—where mental health issues are particularly prevalent (Pedrelli et al., 2015)—and thereby inform tailored approaches to support this population's mental health.

Accordingly, the proposed research seeks to examine the correlation between neuroticism and loneliness, specifically using the Loneliness in Context Questionnaire, and specifically surveying college students. It tests the hypothesis that this correlation is significantly ($\alpha = .001$) greater than $\rho = .00$.

Methods

Participants

Participants will be 400 university students enrolled in a Psychology course who opt to participate in a brief online survey for course credit.

Measures

Demographics. Participants will be asked to indicate their age, gender, and racial and ethnic background. The exact questions and further details are shown in Appendix A.

Big Five Inventory—Neuroticism Subscale. The Big Five Inventory—Neuroticism subscale (John, Donahue, & Kentle, 1991) asks participants to indicate the extent to which they agree or disagree with eight statements about their personal characteristics or thinking and feeling tendencies in day-to-day life (e.g., "I see myself as someone who is depressed, blue;" "I see myself as someone who worries a lot"). The response scale is an ordinal scale ranging from 1 = "Disagree strongly" to 5 = "Agree strongly," with the intermediate response options being 2 = "Disagree a little,"3 = "Neutral; no opinion," and 4 = "Agree a little." The complete scale instructions and items (also indicating which items are reverse scored) are shown in Appendix B.

A composite score is computed by taking the average of the eight items (after appropriate reverse scoring is done).

Loneliness in Context Questionnaire for College Students. The Loneliness in Context Questionnaire for College Students (Asher & Weeks, 2013) asks participants to indicate the frequency with which they feel lonely in various daily or weekly contexts typical for college students. Ten items tap different contexts (e.g., "Class is a lonely place for me;" "I feel sad and alone at social events"), and participants respond on an ordinal scale ranging from 1 = "Never" to 5 = "Always," with the intermediate response options being 2 = "Hardly ever," 3 = "Sometimes," and 4 = "Most of the time."

A composite score is computed by taking the average of the ten items.

Procedure

The online survey will be developed and administered via Qualtrics. Potential participants will be informed about the study via online communications from the university's Psychology Department and may choose to complete the study from their personal device at any time during the current academic semester.

Participants will provide informed consent by indicating that they would like to proceed to the survey questions after reading a brief written explanation (on the first page of the Qualtrics survey) describing what they will be asked to do (i.e., complete an approximately five-minute survey asking them about their personality and mental state) and the possible risks (i.e., experiencing various forms of psychological discomfort while reflecting on oneself and one's feelings) and benefits (i.e., possible improvements in scientific knowledge) of their participation.

Of note, the consent will include a clause indicating that participants' de-identified data (i.e., their responses to all survey questions, including demographic information, but NOT including any information such as IP addresses that could be used to identify them) may be used by other researchers for any purpose and shared on an online, publicly available platform.

Participants who affirm their consent will proceed to the survey questions. All participants will complete the Demographic questions first. To control for possible order effects, participants will be randomized to complete either the Big Five Inventory—Neuroticism subscale or the Loneliness in Context Survey second. The third section will consist of whichever measure participants did not complete second.

Following the survey questions, participants will view a short debriefing description indicating the study aim and providing information about campus mental health resources that may be of aid if participants found that reflecting on their mental state and day-to-day loneliness caused distress or prompted them to think about seeking psychological help.

Data analysis

The following will be used for my analyses (and the writing and producing of this report using the `papaja` template): R (Version 4.5.2; R Core Team, 2025) and the R-packages *apaTables* (Version 2.0.8; Stanley, 2021), *dplyr* (Version 1.1.4; Wickham, François, Henry, Müller, & Vaughan, 2023), *faux* (Version 1.2.3; DeBruine, 2025), *ggplot2* (Version 4.0.1; Wickham, 2016), *groundhog* (Version 3.2.3; Simonsohn & Gruson, 2025), *gtsummary* (Version 2.5.0; Sjoberg, Whiting, Curry, Lavery, & Larmarange, 2021), *papaja* (Version 0.1.4; Aust & Barth, 2025), *psych* (Version 2.5.6; William Revelle, 2025), *readr* (Version 2.1.5; Wickham, Hester, & Bryan, 2024), and *tinylabels* (Version 0.2.5; Barth, 2025). Some code was inspired by (Girard, 2023).

Descriptive statistics will be computed for demographic variables, including mean and standard deviation for age and, for gender and racial and ethnic identities endorsed, counts and proportions.

I will compute Spearman's ρ for the association between the mean neuroticism and loneliness scores and test the hypothesis that this correlation is significantly ($\alpha = .001$) greater than $\rho = .00$ in this sample.

I chose to calculate Spearman's ρ because, although the distributions of simulated data for the mean neuroticism and loneliness scores are both approximately normal (an assumption of Pearson's r; see below for relevant computations and visualizations), neuroticism and loneliness were measured on ordinal scales (indicating Spearman's ρ would be a more appropriate choice; (Schober, Boer, & Schwarte, 2018)).

Results

Using my simulated data, the mean age of participants was 19.71 (SD = 1.31).

Overall, the sample was made up of mostly White women. Descriptive statistics for gender and racial and ethnic identities endorsed are reported in Table 1.

The correlation test revealed that there was not a statistically significant correlation between composite neuroticism and loneliness scores ($\rho = 0.00$; $p = 0.51$). See Figure 3.

Discussion

References

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Figure 1

Distribution of Mean Neuroticism Score

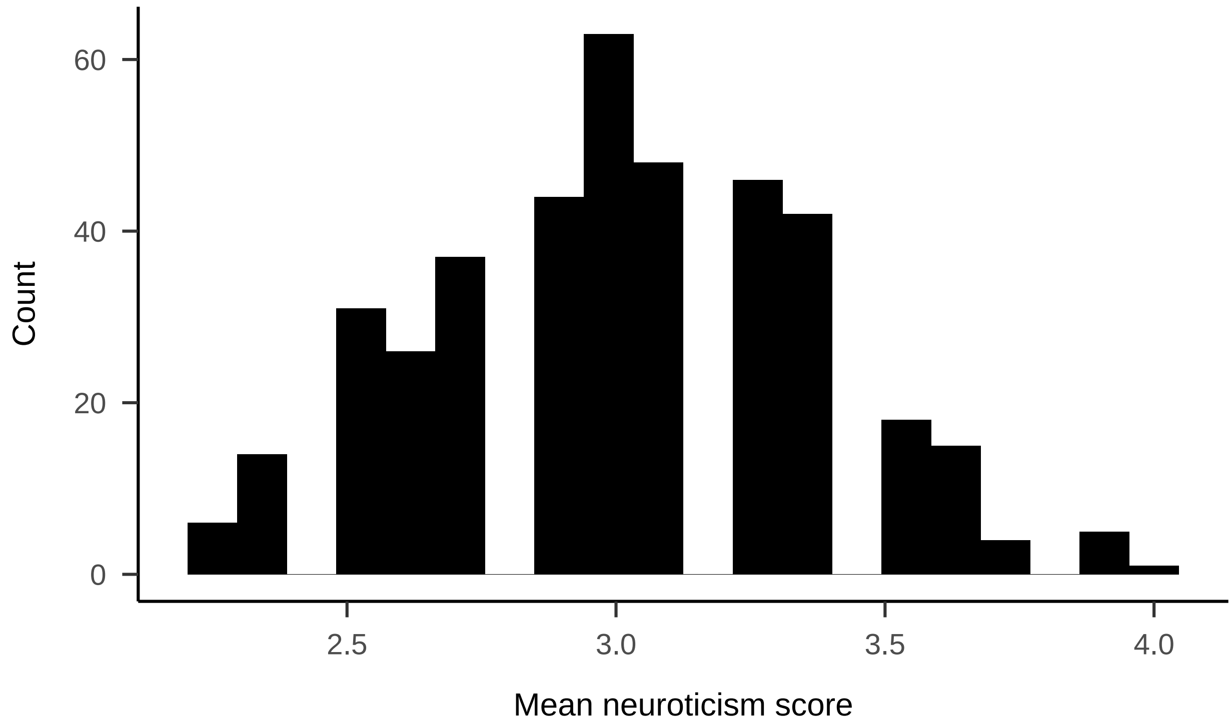


Figure 2

Distribution of Mean Loneliness Score

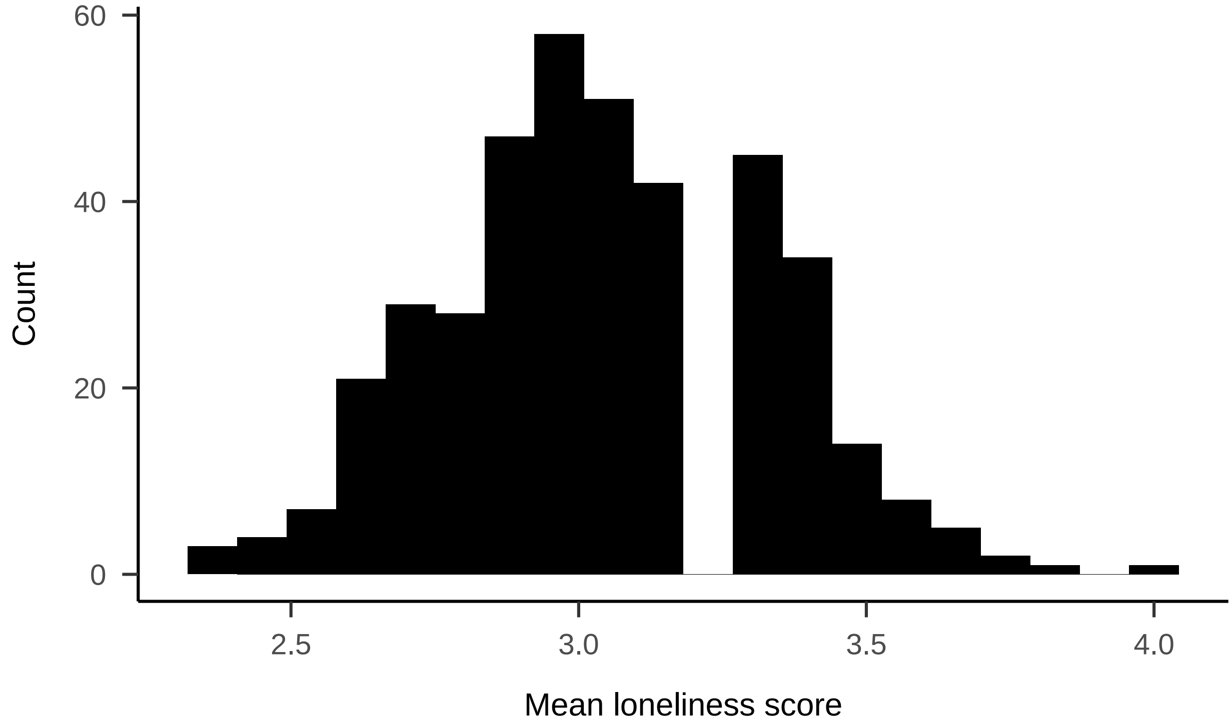
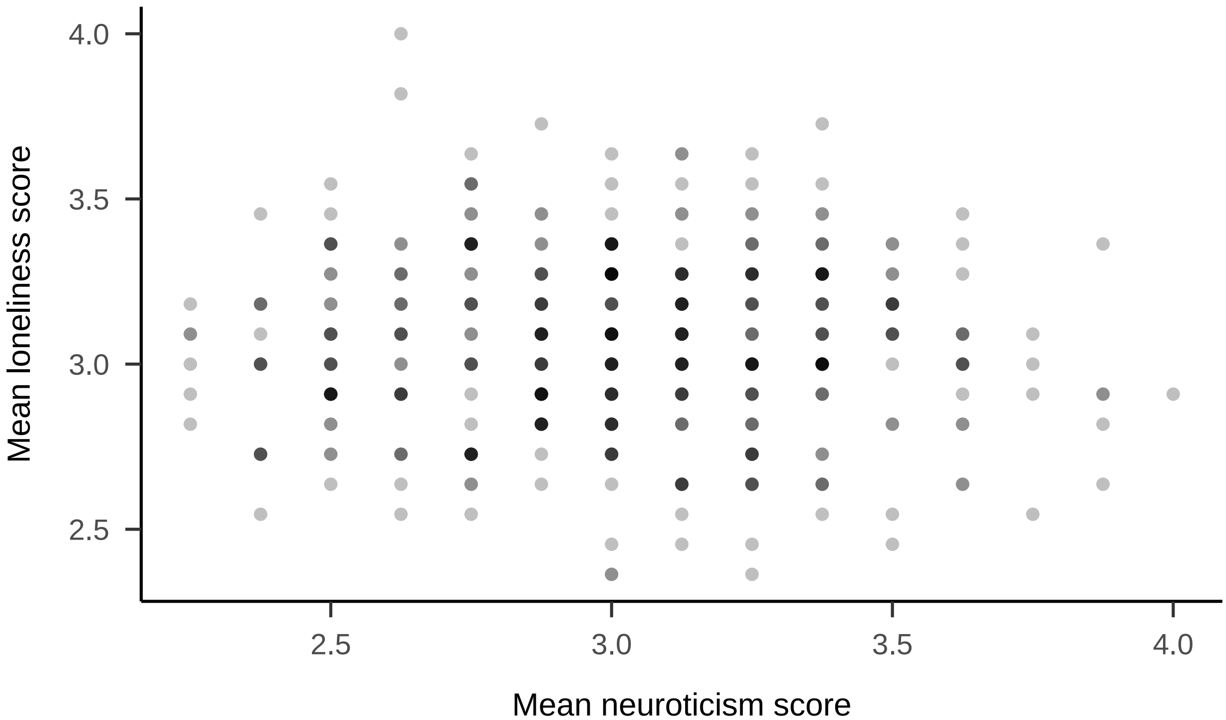


Figure 3

Loneliness vs. Neuroticism (Simulated Data)



Appendix A

Demographics

Participants will be asked the following demographic questions:

What is your current age in years? *(free text response with data validation to ensure that a two-digit integer is entered)*

How would you describe your gender? *(three possible answer choices, of which one may be selected: (1) Man, (2) Woman, (3) Nonbinary or genderqueer)*

How would you describe your racial and ethnic background? Select all that apply. *(seven possible answer choices, of which multiple may be selected: (1) American Indian or Alaska Native, (2) Asian, (3) Black or African American, (4) Hispanic or Latino, (5) Middle Eastern or North African, (6), Native Hawaiian or Pacific Islander, (7) White)*

Appendix B

Big Five Inventory: Neuroticism Subscale (John et al., 1991)

Instructions: In this section, you will read a number of characteristics that may or may not apply to you. For example, do you agree that you are someone who is talkative? Consider how you think and feel in everyday life and indicate how much you agree or disagree with each statement. Remember that there are no right or wrong answers because people think and feel very differently from one another. Just read each question carefully and answer honestly.

I see myself as someone who…

Is depressed, blue

Is relaxed, handles stress well*

Can be tense

Worries a lot

Is emotionally stable, not easily upset*

Can be moody

Remains calm in tense situations*

Gets nervous easily

∗ *Indicates reverse-scored item*

Appendix C

Loneliness in Context Questionnaire for College Students (Asher & Weeks, 2012)

Instructions: Sometimes people can feel lonely in their day-to-day lives. The items in this section ask about your feelings in different contexts. On the scale directly below/beside each item indicate how often you feel the way the item describes. There are no right or wrong answers because people can feel very differently from one another. Just describe how you feel in these contexts (1 = Never, 2 = Hardly Ever, 3 = Sometimes, 4 = Most of the Time, 5 = Always).

Class is a lonely place for me.

I am lonely in the evening.

My place of residence is a lonely place for me.

My free time is a lonely time for me.

I feel sad and alone on weekends.

I am lonely with other people.

I feel sad and alone at social events.

I am lonely during meal times.

I feel sad and alone when I am studying.

Bedtime is a lonely time for me.

Table 1

Demographic Characteristics of Simulated Data

Characteristic	**N = 400**[1]
Gender	
Man	120 (30%)
Nonbinary or genderqueer	18 (4.5%)
Woman	262 (66%)
Racial and ethnic identities endorsed	
American Indian or Alaska Native	9 (2.3%)
Asian	48 (12%)
Black or African American	37 (9.3%)
Hispanic or Latino	55 (14%)
Middle Eastern or North African	24 (6.0%)
Native Hawaiian or Pacific Islander	3 (0.8%)
White	224 (56%)

[1] n (%)